

Find and Fix: Input, Output, and Function Notation

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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| 1. 21 | 5. the corrected input = 3, — |
| 2. the corrected value of $h(-2) = -17$, — | 6. $f(2)$, $g(2)$ and the comparison symbol = $<$, — |
| 3. 4 | 7. $p(5) = 1$, the input x with $p(x) = 5 = 3$ |
| 4. $t(4) = 18$, the input n with $t(n) = 8 = 2$ | 8. $d(6)$ from the rule = 20, — |
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What is verified

4 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 2, 5, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.F.A – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

2. If they got -38 : put the input where the constant belongs and the constant where the input belongs, computing $4(-9) - 2$
3. If they got 37: treated 15 as the input and evaluated $f(15)$ instead of solving for x
5. If they got 19: repeated the output 19 instead of naming the input that produces it
7. If they got 1: substituted 5 for x instead of solving, answering with $p(5)$
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1. $f(x) = 5x + 1$. Find $f(4)$.

Solution: The 4 inside the brackets is the input, so it replaces x : $f(4) = 5(4) + 1 = 20 + 1$. $f(4) = 21$

2. Devin wrote $h(-2) = 4(-9) - 2 = -38$ for $h(x) = 4x - 9$. Correct it and say what he did wrong.

(a) Solution: The input is -2 , so -2 replaces x and the -9 stays where it is: $h(-2) = 4(-2) - 9 = -8 - 9$. $h(-2) = -17$

(b) Diagnosis (instructor-judged). A correct response says the two numbers were swapped: Devin multiplied 4 by the constant -9 and then subtracted the input -2 , when the input belongs in the x slot and the constant stays at the end. Accept any wording that names the swap. Correcting the value without saying what went wrong is only half the task.

3. $f(x) = 2x + 7$. Find the value of x for which $f(x) = 15$.

Solution: Here 15 is the *output*, so it goes on the left of the equals sign and x stays unknown. Solve $2x + 7 = 15$:

$$2x = 8, \quad x = \frac{8}{2}$$

Check by substituting back: $2(4) + 7 = 15$ ✓.

$x = 4$

4. From the table for t : find $t(4)$, and find the input n with $t(n) = 8$.

Solution: The two questions travel through the table in opposite directions. For $t(4)$, find 4 in the top row and read down: the output is 18. For $t(n) = 8$, find 8 in the bottom row and read up: the input is 2.

$t(4) = 18$	$n = 2$
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5. Rosa was asked which input of c gives an output of 19, and answered 19. Correct her and say what went wrong.

(a) Solution: Find 19 in the output row of the table and read up to the input row above it. The input paired with 19 is 3.

$x = 3$

(b) Diagnosis (instructor-judged). A correct response says that 19 is an output — it lives in the bottom row — and the question asked for the input in the top row that produces it. Accept any wording that separates the two rows or the two roles. Simply writing the corrected number is not a diagnosis.

6. $f(x) = 3x + 4$, $g(x) = 4x + 3$. Compare $f(2)$ and $g(2)$, and judge the claim that they must be equal.

Solution: $f(2) = 3(2) + 4 = 6 + 4 = 10$ and $g(2) = 4(2) + 3 = 8 + 3 = 11$, so the first is smaller.

$10 < 11$

Judgement (instructor-judged). A correct response says the claim is wrong and explains why swapping matters: only the coefficient is multiplied by the input, so moving the 3 and the 4 changes which number gets doubled. Accept any wording that ties the difference to the coefficient. A bare “no” is not an explanation.

7. $p(x) = -2x + 11$. (a) Find $p(5)$. (b) Find x with $p(x) = 5$.

Solution: In (a) the 5 is an input, so substitute it: $p(5) = -2(5) + 11 = -10 + 11$. In (b) the 5 is an output, so it goes on the right of the equation and you solve: $-2x + 11 = 5$, so $-2x = -6$ and $x = 3$. The same number plays opposite roles in the two parts.

$p(5) = 1$	$x = 3$
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8. $d(x) = 3x + 2$ and $d(6) = 20$. Rewrite Kofi's sentence correctly, then check the statement with the rule.

(a) Rewrite (instructor-judged). A correct sentence puts 6 in the input role and 20 in the output role: “when the input is 6, the output is 20”. Accept any equivalent wording (“ d takes 6 to 20”, “an input of 6 gives an output of 20”). Kofi has the two roles the wrong way round, so his sentence describes a different pairing entirely.

(b) Solution: Substitute the input into the rule: $d(6) = 3(6) + 2 = 18 + 2$, which confirms the statement is true. $d(6) = 20$